Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Chemistry Section A

Section Id :	69527829
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527829
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 695278351 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

How many grams of residue is obtained by heating 2.76 g of silver carbonate?
(Given : Molar mass of C, O and Ag are 12, 16 and 108 g mol^{-1} respectively)

Options :

6952781191. 1.08 g

6952781192. 2.16 g

6952781193. 3.24 g

6952781194. 4.32 g

Question Number : 52 Question Id : 695278352 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Arrange the following atomic orbitals of multi electron atoms in order of increasing energy.

A. $n = 3, l = 2, m = +1$
B. $n = 4, l = 0, m = 0$
C. $n = 6, l = 1, m = 0$
D. $n = 5, l = 1, m = +1$
E. $n = 2, l = 1, m = +1$

Choose the correct answer from the options given below:

Options :

6952781195. C < D < B < A < E

6952781196. B < A < E < C < D

6952781197. E < C < D < B < A

6952781198. $E < B < A < D < C$

Question Number : 53 Question Id : 695278353 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the **correct** statements from the following :

A. Heisenberg uncertainty principle is applicable to electrons.
B. The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.
C. The energy of 2s orbital of H atom is equal to the energy of 2s orbital of Li.
D. The electronic configuration of Cr is [Ar] $3d^5\ 4s^1$

Choose the correct answer from the options given below:

Options :

6952781199. A, B and C Only

6952781200. A, B and D Only

6952781201. B, C and D Only

6952781202. A, C and D Only

Question Number : 54 Question Id : 695278354 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

What is the mole fraction of water in 10% by weight (w/w) of aqueous urea solution?

[Given: Molar mass of H, O, C and N are 1, 16, 12 and 14 g mol^{-1} respectively.]

Options :

6952781203. 0.825

6952781204. 0.032

6952781205. 0.867

6952781206. 0.967

Question Number : 55 Question Id : 695278355 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

M_3A_2 is a sparingly soluble salt of molar mass y g mol^{-1} and solubility x g L^{-1}.

The ratio of the molar concentration of the anion (A^{3-}) to the solubility product of the salt is

Options :

6952781207. $\frac{1}{54}\cdot\frac{y^4}{x^4}$

6952781208. $\frac{y^5}{108x^4}$

6952781209. $108 \cdot \frac{x^5}{y^5}$

6952781210. $\frac{1}{108}\frac{y^4}{x^4}$

Question Number : 56 Question Id : 695278356 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Arrange the following resultant mixtures in increasing order of their pH values

A. 10 mL 0.2 M $Ca(OH)_2$ + 25 mL 0.1 M HCl
B. 10 mL 0.01 M H_2SO_4 + 10 mL 0.01 M $Ca(OH)_2$
C. 10 mL 0.1 M H_2SO_4 + 10 mL 0.1 M KOH

Choose the correct answer from the options given below:

Options :

6952781211. $B < C < A$

6952781212. $C < A < B$

6952781213. $C < B < A$

6952781214. $A < C < B$

Question Number : 57 Question Id : 695278357 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

First order gas phase reaction

$A \rightarrow B + C$

p_i = initial pressure of gas A, p_t = total pressure of the reaction mixture at time t

Expression of rate constant (k) is

Options :

6952781215. $\frac{1}{t}\ln\frac{p_i}{2p_i - p_t}$

6952781216. $\frac{1}{t}\ln\frac{2p_i}{p_i - p_t}$

6952781217. $\frac{1}{t}\ln\frac{p_i}{3p_i - 2p_t}$

6952781218. $\frac{1}{t}\ln\frac{3p_i}{4p_i - p_t}$

Question Number : 58 Question Id : 695278358 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: The correct order of electronegativity of fluorine, oxygen and nitrogen is F > O > N.

Statement II: The oxidation state of oxygen in OF_2 is +2 and in Na_2O is −2.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952781219. Both Statement I and Statement II are true

6952781220. Both Statement I and Statement II are false

6952781221. Statement I is true but Statement II is false

6952781222. Statement I is false but Statement II is true

Question Number : 59 Question Id : 695278359 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Correct statements from the following are:

A. Nitrogen in oxidation states from +1 to +4 disproportionates in acid medium.
B. Nitrogen has the ability to form dπ - pπ multiple bonds with itself and other elements with small size and high electronegativity.
C. N-N single bond is stronger than P-P single bond.
D. Nitrogen has highest density in its group due to small size.
E. The maximum covalency of nitrogen is four since it has only four valence orbitals for bonding.

Choose the correct answer from the options given below:

Options :

6952781223. B, C and D Only

6952781224. C, D and E Only

6952781225. A, C and E Only

6952781226. A and E Only

Question Number : 60 Question Id : 695278360 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following is **NOT** a physical or chemical characteristics of interstitial compounds?

Options :

6952781227. They have high melting points, higher than those of pure metals.

6952781228. They are very soft and ionic in nature.

6952781229. They retain metallic conductivity.

6952781230. They are chemically inert and usually non-stoichiometric.

Question Number : 61 Question Id : 695278361 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The **correct** statements about metal carbonyls are

A. The metal-carbon bonds in metal carbonyls possess both σ and π-character.
B. Due to synergic bonding interactions between metal and CO ligand, the metal-carbon bond becomes weak.
C. The metal-carbon σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of metal.
D. The metal-carbon π bond is formed by the donation of electrons from filled d-orbital of metal into vacant π* orbital of CO.

Choose the correct answer from the options given below:

Options :

6952781231. A and B Only

6952781232. A, C and D Only

6952781233. B and C Only

6952781234. A and D Only

Question Number : 62 Question Id : 695278362 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: Each electron in e_g orbitals destabilizes the orbitals by $+0.6\ \Delta_o$ and each electron in the t_{2g} orbitals stabilizes the orbitals by $-0.4\ \Delta_o$ in an octahedral field on the basis of crystal field theory.

Statement II: All the d - orbitals of the transition metals have the same energy in their free atomic state but when a complex is formed the ligands destroy the degeneracy of these orbitals on the basis of crystal field theory.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952781235. Both Statement I and Statement II are correct

6952781236. Both Statement I and Statement II are incorrect

6952781237. Statement I is correct but Statement II is incorrect

6952781238. Statement I is incorrect but Statement II is correct

Question Number : 63 Question Id : 695278363 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: On the basis of inductive effect, the order of stability of alkyl carbanions is $CH_3^- > CH_3-CH_2^- > (CH_3)_2CH^- > (CH_3)_3C^-$.

Statement II: Allyl and benzyl carbanions are more stabilised by inductive effect and not by resonance effect.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952781239. Both Statement I and Statement II are correct

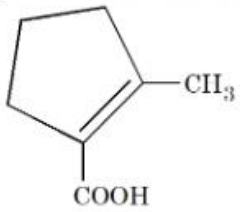
6952781240. Both Statement I and Statement II are incorrect

6952781241. Statement I is correct but Statement II is incorrect

6952781242. Statement I is incorrect but Statement II is correct

Question Number : 64 Question Id : 695278364 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

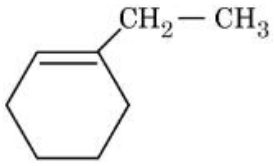
"P" is a hydrocarbon of molecular formula:- C_8H_{14}. On ozonolysis, "P" forms "Q". "Q" on treatment with alkali under reflux condition produces "R", which on treatment with I_2/ NaOH gives a yellow precipitate. Acidification of the solution gives "S". The structure of "S" is given below:-

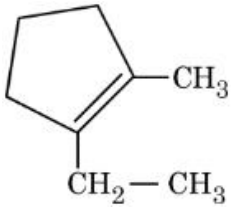


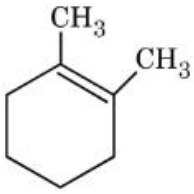
The correct structure of "P" is

Options :

6952781243. 

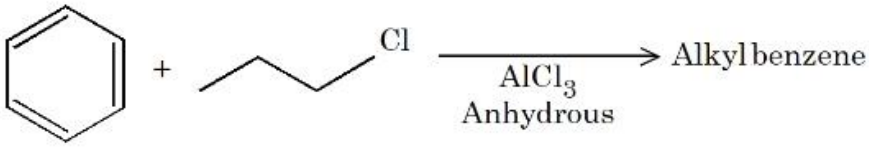
6952781244. 

6952781245. 

6952781246. 

Question Number : 65 Question Id : 695278365 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For the following Friedel Craft's alkylation reaction, which of the statements are **correct**?



A. Major product is n-propyl benzene.
B. iso-propyl carbocation intermediate is also generated.
C. Multiple substitution is inevitable.
D. Introducing electron-donating substituent on benzene will not produce any alkyl benzene.

Choose the correct answer from the options given below:

Options :

6952781247. A and D only

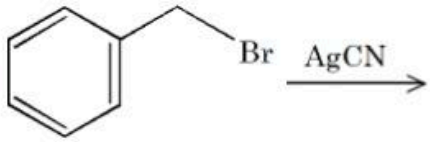
6952781248. B and C only

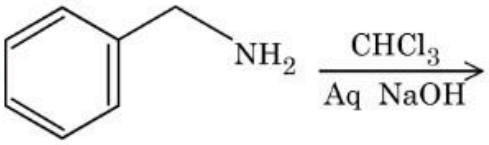
6952781249. A and C only

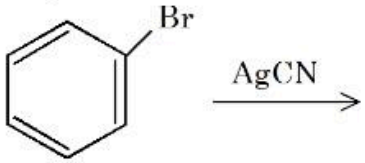
6952781250. B and D only

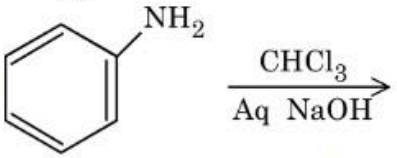
Question Number : 66 Question Id : 695278366 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

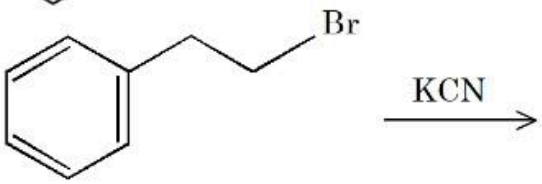
Benzyl isocyanide can be obtained from

A. 

B. 

C. 

D. 

E. 

Choose the correct answer from the options given below:

Options :

6952781251. A and B Only

6952781252. A and C Only

6952781253. B and D Only

6952781254. D and E Only

Question Number : 67 Question Id : 695278367 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider compounds A, B and C with following structural formulae

A = $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - OH$

B = $CH_2 = CH - CH_2 - CH_2 - CH_3$

C = $HO - CH_2 - CH_2 - CH(OH) - CH_3$

For the conversion of B from A, reagent (D) required is _____ and structural formula of product (E) obtained when C undergoes same reaction using excess reagent (D) is _____.

Options :

6952781255.

D	E
Conc. H_2SO_4	$CH_2 = CH - CH(OH)CH_3$

6952781256.

D	E
PCC	$HO - CH_2 - CH_2 - CH = CH_2$

6952781257.

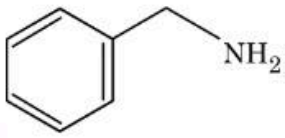
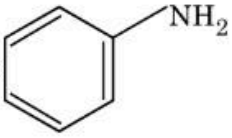
D	E
PCC	$CH_2 = CH - CH = CH_2$

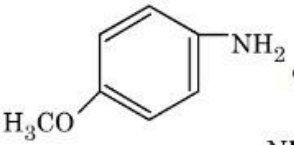
6952781258.

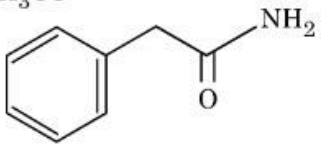
D	E
Conc. H_2SO_4 or H_3PO_4	$CH_2 = CH - CH = CH_2$

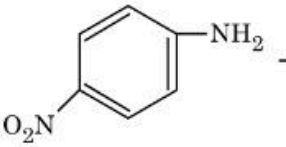
Question Number : 68 Question Id : 695278368 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the **incorrect** statements.

A.  is a stronger base than 

B.  can be synthesized by Gabriel phthalimide synthesis.

C.  $\xrightarrow{Br_2,\ NaOH}$ primary aromatic amine

D.  $\xrightarrow[(ii)\ \Delta]{(i)\ NaNO_2,\ HCl, 0°C}$ product will dissolve in NaOH

Choose the correct answer from the options given below:

Options :

6952781259. A and D Only

6952781260. A and C Only

6952781261. B and C Only

6952781262. A and B Only

Question Number : 69 Question Id : 695278369 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the **correct** statements.

A. Glucose exists in two anomeric forms.
B. Anomers of glucose differ in configuration at C−1 in cyclic hemiacetal structure.
C. Melting point of α - anomer of glucose is greater than β - anomer.
D. Specific rotation of α - anomer is +19° while for β - anomer is +112°
E. α and β - anomers of glucose are prepared by crystallization of saturated glucose solution at 303 K and 371 K respectively.

Choose the correct answer from the options given below:

Options :

6952781263. A and B Only

6952781264. B and C Only

6952781265. A, B and D Only

6952781266. A, B and E Only

Question Number : 70 Question Id : 695278370 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements:

Statement I: Sodium dichromate and potassium dichromate are classified as primary standards in titrimetric analysis.

Statement II: Phenolphthalein is a weak base, therefore it dissociates in acidic medium.

In the light of the above statements, choose the ***correct*** answer from the options given below

Options :

6952781267. Both Statement I and Statement II are ture

6952781268. Both Statement I and Statement II are false

6952781269. Statement I is true but Statement II is false

6952781270. Statement I is false but Statement II is true

Chemistry Section B

Section Id :	69527830
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527830
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Id : 695278371 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following species:

BrF_5, XeF_5^-, BF_4^-, ICl_4^-, XeF_4, SF_4, NH_4^+, ClF_3, XeF_2, ICl_2^-

Number of species having sp^3d hybridized central atom is _____.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 695278372 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

In an estimation of sulphur by Carius method 0.2 g of the substance gave 0.6 g of $BaSO_4$. The percentage of sulphur in the substance is _____%.

(Given molar mass in g mol^{-1} S : 32, $BaSO_4$: 231)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 73 Question Id : 695278373 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

One mole of phenol is treated with dilute HNO_3 at 298 K to give a mixture of products. The mixture is separated by steam distillation. The steam volatile compound (X) is separated. The increase in percentage of oxygen in (X) with respect to phenol is _____$\times 10^{-1}$ %

(Given molar mass in g mol^{-1} H:1, C:12, N:14, O:16)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 74 Question Id : 695278374 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The values of pressure equilibrium constant recorded at different temperatures for the following equilibrium reaction have been given below

$A(g) \rightleftharpoons B(g) + C(g)$

$\frac{1}{T}(K^{-1})$	$\log_{10}K_p$
0.05	3.5
0.06	2.5
0.07	1.5

The magnitude of $\frac{\Delta H^\circ}{R}$ calculated from the above data is _____. (Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 75 Question Id : 695278375 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If the half life of a first order reaction is 6.93 minutes then the time required for completion of 99% of the reaction will be _____ minutes .
(Given : log 2 = 0.3010)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1